UNIVERSITY OF TUNIS
TUNIS BUSINESS SCHOOL
Business Analytics Program

DATA MINING PROJECT

US Airline Twitter Sentiment Analysis

Course	Data Mining
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Program	Bachelor in Business Analytics – Junior Year
Semester	Spring 2025–2026
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Dataset link	https://www.kaggle.com/datasets/zahranusratt/us-airline-twitter-sentiment-analysis-dataset
Python Script	https://colab.research.google.com/drive/19LRH3418FxDexw2SALluxwSgUGXDKX5X?usp=sharing

1. Problem Statement & Motivation

The airline industry is highly customer-facing, with passengers frequently sharing real-time feedback on Twitter. These tweets provide valuable but underused business insights. However, airlines often lack automated tools to analyze this large volume of unstructured data effectively. Manual analysis is slow and subjective, while simple methods miss the nuances of language. The key challenge is to accurately classify tweet sentiment (negative, neutral, positive) and uncover hidden themes without relying on predefined labels.

1.1 Research Questions

- Can a machine learning model reliably classify the sentiment of airline-related tweets into three categories: negative, neutral, and positive?
- Does addressing class imbalance through balanced class weighting significantly improve classification performance for minority sentiment classes?
- Can unsupervised clustering (K-Means) recover meaningful sentiment groupings from raw TF-IDF representations without using any labels?
- How do the supervised and unsupervised approaches compare in terms of accuracy, robustness, and practical utility?

1.2 Target Variable

The target variable is airline_sentiment, a three-class categorical label annotated by human reviewers. It takes one of three values: negative (62.7% of observations), neutral (21.2%), or positive (16.1%). This pronounced class imbalance is a central methodological challenge that motivates the use of balanced class weighting in the supervised model.

1.3 Analytical Objectives

- Apply NLP techniques to raw tweets to convert them into a structured, machine-readable format.
- Train and evaluate a Logistic Regression model using TF-IDF, with and without class weighting to handle imbalance.
- Use PCA to reduce TF-IDF dimensionality for visualization and clustering.
- Apply K-Means clustering to uncover latent groups, evaluated using silhouette score and purity.
- Compare supervised and unsupervised methods to gain complementary insights into sentiment patterns.

2. Dataset Description

The dataset used in this project is the Twitter US Airline Sentiment dataset from Kaggle, a well established benchmark in text mining and natural language processing. It consists of tweets collected over a one-week period in February 2015, which were manually annotated by contributors as negative, neutral, or positive. For tweets labeled as negative, the primary cause of dissatisfaction—such as customer service issues, flight delays, or cancellations—was also identified. Each observation represents a single tweet directed at one of six major US airlines: United Airlines, American Airlines, Southwest Airlines, Delta Air Lines, US Airways, and Virgin America, for a total of 14,640 labeled instances.

2.1 Key Variables

Variable	Type	Description
text	String	Raw tweet content posted by the passenger
airline_sentiment	Categorical (3 classes)	Overall sentiment label: negative, neutral, positive
negativereason	Categorical (11 categories)	Reason for negative sentiment (if applicable)
airline	Categorical (6 values)	Airline mentioned in the tweet
name	String	Twitter username of the poster
retweet_count	Integer	Number of retweets the tweet received
tweet_coord	String / Null	Geographic coordinates of the tweet (mostly missing)
tweet_created	Datetime	Timestamp of tweet creation

3. Data Preparation

Raw social media text requires extensive preprocessing before any machine learning model can be applied.

3.1 Column Reduction

Five columns were removed from the original dataset before any further processing:

- tweet_id: A unique numerical identifier with no predictive or descriptive value.
- tweet_location and user_timezone: Both fields had extensive missing data (over 30% null) and were not required for sentiment classification.
- airline_sentiment_confidence and negativereason_confidence: These confidence scores describe the annotation quality rather than the content of the tweet, and including them would introduce label leakage.

3.2 Text Preprocessing Pipeline

Each tweet was processed through a seven-step pipeline before feature extraction:

Step	Operation	Justification
1	Remove @mentions	Airline handle tags add noise; airline identity is captured separately
2	Remove URLs	Links carry no sentiment information
3	Remove punctuation	Reduces vocabulary sparsity without semantic loss
4	Lowercase	Normalises tokens (e.g., 'GREAT' = 'great')
5	Tokenise	Splits text into individual words
6	Remove stopwords	Custom list: NLTK English stops + airline names + 22 domain-specific tokens
7	Length filter (>2 chars)	Removes uninformative single and double character tokens

3.3 Label Encoding

The categorical sentiment labels were mapped to integer codes for compatibility with scikit-learn: negative → 0, neutral → 1, positive → 2. The original text labels were preserved in a separate column for interpretability.

3.4 Train / Test Split

The dataset was split into a training set (80%, 11,712 tweets) and a test set (20%, 2,928 tweets) using stratified sampling (stratify=y, random_state=42). Stratification ensures that the class proportions are preserved in both splits, which is essential given the significant class imbalance.

3.5 Feature Extraction: TF-IDF

Term Frequency – Inverse Document Frequency (TF-IDF) was used to convert the cleaned tweet text into a numerical feature matrix. The following configuration was applied:

- sublinear_tf=True: Replaces raw term frequency tf with 1 + log(tf). This dampens the dominance of very frequent terms and produces more balanced feature vectors.
- min_df=2: Tokens appearing in only one tweet are treated as noise and excluded. This also reduces the vocabulary size.
- max_df=0.95: Tokens present in 95% or more of tweets are excluded as they provide no discriminative value.

The resulting TF-IDF vocabulary contained approximately 3,700 terms. The training matrix was of shape (11,712 × 3,700) and the test matrix (2,928 × 3,700).

4. Exploratory Data Analysis

Before building any models, a thorough exploratory analysis was conducted to understand the distribution of sentiment across airlines and to identify the dominant themes in each sentiment class.

4.1 Overall Sentiment Distribution

The overall distribution of sentiment across all 14,640 tweets is shown in Figure 4.1. Negative tweets represent a large majority (9,178 tweets, 62.7%), followed by neutral (3,099 tweets, 21.2%) and positive (2,363 tweets, 16.1%). This skewed distribution confirms that airline-directed social media traffic is predominantly complaint-driven, and it directly motivates the use of class-balanced weighting in the supervised model.

4.2 Sentiment Distribution by Airline

As shown in Figure 4.2, Virgin America stands out as the airline with the highest proportion of positive sentiment, while United Airlines and US Airways concentrate the largest share of negative feedback. This cross-airline variation suggests that sentiment is not uniform across carriers, and that airline-specific service quality is a meaningful driver of customer opinion.

4.3 Negative Complaint Reasons

Among negative tweets, Figure 4.3 reveals the most frequently cited reasons for dissatisfaction. 'Customer Service Issue' is by far the most common complaint category, accounting for the largest single share of negative tweets. 'Late Flight' and 'Cancelled Flight' follow as the second and third most cited reasons. This finding has direct practical implications for airline management: improvements in customer-facing service quality and on-time performance would address the vast majority of negative social media feedback.

4.4 Word Clouds by Sentiment Class

Figure 4.4 displays word clouds generated from the cleaned tweet text, stratified by sentiment class. Negative tweets are dominated by operational failure keywords such as 'flight', 'hours', 'cancelled', 'bag', and 'service'. Neutral tweets tend to contain informational or transactional language. Positive tweets feature appreciative vocabulary such as 'great', 'thanks', 'love', and 'amazing'. These distinct vocabularies confirm that TF-IDF-based feature extraction should be able to discriminate between sentiment classes meaningfully.

5. Methods and Analysis

5.1 Supervised Method: Logistic Regression with TF-IDF

Logistic Regression is a well-established baseline classifier for text categorisation tasks. It is computationally efficient, interpretable, and performs competitively on high-dimensional sparse feature matrices such as those produced by TF-IDF. Its probabilistic framework also makes it suitable for multi-class problems through the multinomial softmax formulation.

5.1.1 Handling Class Imbalance

The dataset's unbalanced class split creates a strong incentive for the model to predict the majority class (negative) whenever uncertain. Without correction, this results in acceptable raw accuracy but poor recall on neutral and positive classes. The class_weight='balanced' option addresses this by inversely weighting each class proportional to its frequency.

The computed weights are:

Sentiment Class	Label	Computed Weight
Negative	0	0.529 (majority — downweighted)
Neutral	1	1.574 (upweighted ×1.57)
Positive	2	2.067 (upweighted ×2.07)

5.2 Unsupervised Model — K-Means Clustering

K-Means clustering was applied to assess whether sentiment-related structure exists in the tweet feature space without any label information — reflecting real-world scenarios where labelled data may not be available.

5.2.1 Dimensionality Reduction via PCA

K-Means suffers from the curse of dimensionality in high-dimensional sparse spaces. Principal Component Analysis (PCA) was applied to reduce the 12,000-dimensional TF-IDF matrix to 50 principal components before clustering. These 50 components captured approximately 14.1% of the total variance — a seemingly low figure that is, in fact, typical for very sparse, high-dimensional text data where no single direction concentrates much variance.

5.2.2 Optimal K Selection

The Within-Cluster Sum of Squares (WCSS) was computed for K = 2 to 10 using the Elbow Method A clear inflection point appeared at K = 3, naturally matching the three sentiment classes. K = 3 was therefore selected.

5.2.3 Cluster Evaluation

Since K-Means is unsupervised, two metrics were used to evaluate the quality of clusters:

- **Silhouette Score:** Silhouette Score — measures cohesion vs. separation. Values near 1 indicate tight, well-separated clusters; values near 0 indicate overlapping clusters.
- **Cluster Purity:** Cluster Purity — for each cluster, the proportion of tweets belonging to the dominant true label. Higher purity means the cluster is dominated by a single sentiment class.

6. Results and Interpretation

6.1 Supervised Model: Logistic Regression + TF-IDF

6.1.1 Before vs. After Class Weighting

Table 6.1 summarises the performance metrics before and after applying balanced class weighting.

(check figure 6.1.1)

Metric	Before Weighting	After Weighting	Change
Accuracy	0.7725	0.7490	-0.0236 (↓ Dropped)
Balanced Accuracy	0.6462	0.7260	+ 0.0798 (↑ Improved)
Macro F1	0.6813	0.7063	+ 0.0250 (↑ Improved)
Negative F1	0.8589	0.8285	-0.0304 (↓ Dropped)
Neutral F1	0.5217	0.5966	+0.0748 (↑ Improved)
Positive F1	0.6633	0.6939	+0.0306 (↑ Improved)

- The results show an important insight: if we only consider accuracy, the unweighted model seems slightly better (77.25% vs. 74.90%). However, this is misleading because it mainly favors the majority class (Negative) and does not perform as well on Neutral and Positive tweets.

- After applying class weighting, the model becomes more balanced. Balanced Accuracy increases by about **8%** , and Macro F1 improves slightly. In addition, both Neutral and Positive F1 scores improve, meaning the model is better at detecting these classes.

- Although the Negative F1 score and overall accuracy decrease slightly, this is a reasonable trade-off since the model was already strong at predicting negative tweets.

6.1.2 Confusion Matrix Analysis

The confusion matrices (Figure 6.1.2) show that before weighting, the model misclassifies a large proportion of Neutral and Positive tweets as Negative. After weighting, the off-diagonal mass shifts substantially: Neutral and Positive tweets are correctly identified much more frequently.

6.2 PCA Results

The cumulative explained variance curve for PCA applied to the TF-IDF matrix is shown in Figure 6.2.1) 50 principal components retain approximately 15.8% of the total variance, while 100 components retain approximately 23.7%. The slow accumulation of variance confirms the highly distributed nature of information in text data: no small set of components can summarise the full semantic space.

Figure 6.2.2 presents the 2D PCA scatter plot of the training set, coloured by true sentiment label. The three classes overlap substantially along both principal components, which explains why unsupervised clustering will face more difficulty than the supervised model. However, a visible gradient from negative (left/bottom) to positive (right/top) is discernible along PC1, suggesting that the first principal component captures a polarity axis.

6.3 K-Means Clustering Results

6.3.1 Quantitative Evaluation

The K-Means model (K=3, trained on PCA-50 features) achieved the following scores:

Metric	Value	Interpretation
Cluster Purity	0.6651	66.5% of training tweets fall in the dominant-class cluster
Silhouette Score	0.2179	Clusters are weakly separated but statistically coherent

A silhouette score of 0.22 is common in text clustering tasks, where overlapping language and high-dimensional representations make clear separation difficult.(figure 6.3.1)

In this context, a value above 0.2 is generally considered acceptable, indicating that the clustering captures a meaningful—though not perfectly distinct—structure in the data.

6.3.2 Cluster Profiles

The three clusters were profiled by their dominant sentiment class (figure 6.3.2):

- Cluster 0: Dominated by Negative tweets, with high internal purity. Captures the most operationally-focused complaints.
- Cluster 1: A mixed cluster, primarily capturing Negative and Neutral boundary cases where the complaint language is less extreme.
- Cluster 2: Aligned with Positive tweets, though this cluster has the lowest purity due to the relatively small size of the positive class.

6.4 Supervised vs. Unsupervised Comparison

Criterion	LR + TF-IDF (Weighted)	K-Means (K=3, PCA-50)
Accuracy / Purity	0.7490	0.6651
Balanced Accuracy	0.7260	N/A
Macro F1	0.7063	N/A
Silhouette Score	N/A	0.2179
Requires Labels	Yes	No
Class Imbalance Handled	Yes (balanced weights)	N/A

- The supervised model performs better on all accuracy metrics, as it is trained using labeled data.
- K-Means, without labels, still captures about two-thirds of the sentiment structure from the tweets.
- The two methods are complementary: Logistic Regression classifies tweets, while K-Means explores underlying patterns without needing labels.

7. Conclusion

7.1 Summary of the Project

This project applied data mining and natural language processing techniques to the Twitter US Airline Sentiment dataset, comprising 14,640 tweets directed at six major US airlines collected in February 2015. The central objective was to build and evaluate sentiment classification and clustering models capable of automatically distinguishing negative, neutral, and positive airline customer opinions.

7.2 Methods Used

- Text Mining and NLP: tweet cleaning, tokenisation, custom stopword removal, and TF-IDF feature extraction.
- Supervised Learning: Logistic Regression with balanced class weighting, evaluated using accuracy, balanced accuracy, and per-class F1 scores.
- Dimensionality Reduction: PCA applied to the TF-IDF matrix for clustering and 2D visualisation.
- Unsupervised Learning: K-Means clustering (K=3 selected via the Elbow Method), evaluated with silhouette score and cluster purity.

7.3 Key Results

- The weighted Logistic Regression model achieved an accuracy of 74.9%, a balanced accuracy of 72.6%, and a macro F1 score of 70.6%.
- Balanced class weighting produced gains for minority classes: Positive F1 increased by +3.1 percentage points, and Neutral F1 by +7.5 percentage points.
- K-Means (K=3, PCA-50) achieved a cluster purity of 66.5% and a silhouette score of 0.218, confirming that meaningful sentiment structure exists in the tweet space even without labels.
- PCA revealed that 50 principal components capture approximately 15.8% of variance, reflecting the high-dimensional, distributed nature of text feature spaces.

7.4 Practical Insights

From a business perspective, most negative feedback is driven by Customer Service Issues, Late Flights, and Cancelled Flights, making these key areas for improvement. Virgin America stands out with a more positive sentiment profile, serving as a useful benchmark for other airlines.

From a methodological perspective, the results show that evaluation metrics matter in imbalanced datasets. Accuracy alone can be misleading, so balanced accuracy and macro F1 should be prioritized for a more reliable assessment of model performance.

7.5 Limitations

Several limitations should be acknowledged. The dataset was collected over a single week in 2015 and may not reflect current airline-passenger dynamics. The feature space is restricted to word-level unigrams; incorporating bigrams, trigrams, or contextual embeddings could substantially improve classification performance. Additionally, the manual annotation of sentiment labels introduces subjectivity, and annotator disagreements could affect model quality.

Appendix — Figures

All figures referenced in the body of the report are presented below in order of appearance.

Figure 4.1 — Overall Sentiment Distribution

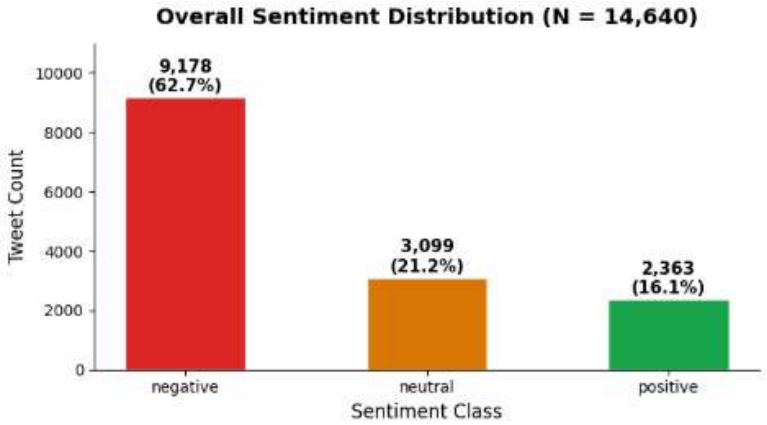


Figure 4.2 — Sentiment Distribution by Airline (%)

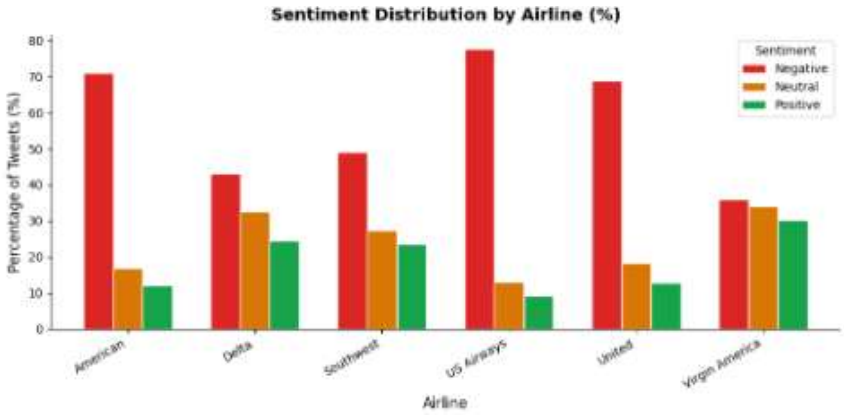


Figure 4.3 — Negative Complaint Reasons

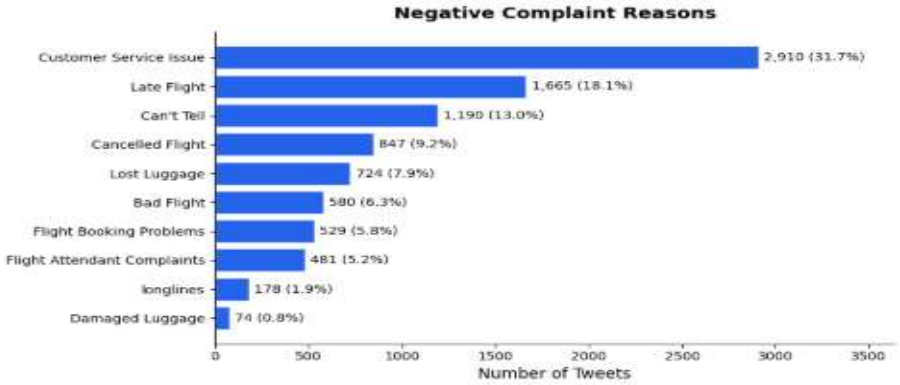


Figure 4.4 — Word Clouds by Sentiment Class



Figure 6.1.1 — Class Weighting Impact: Per-Metric Comparison

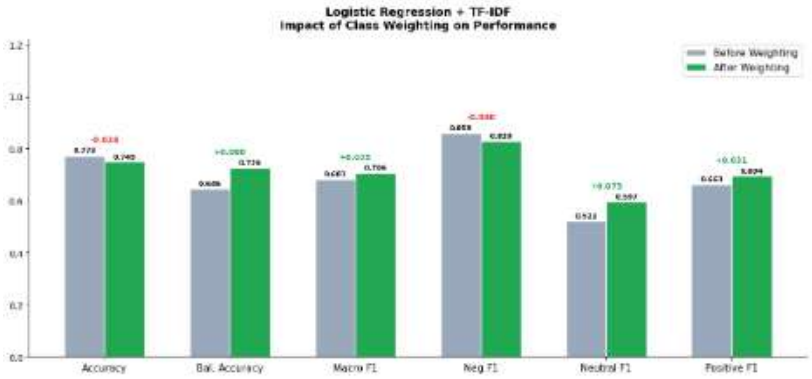


Figure 6.1.2 — Confusion Matrices: Before vs. After Class Weighting

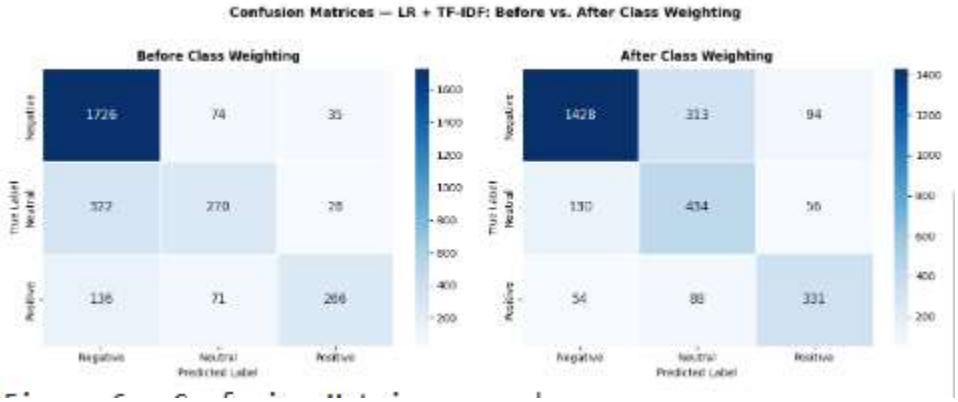


Figure 6.2.1 — Cumulative Explained Variance — PCA on TF-IDF

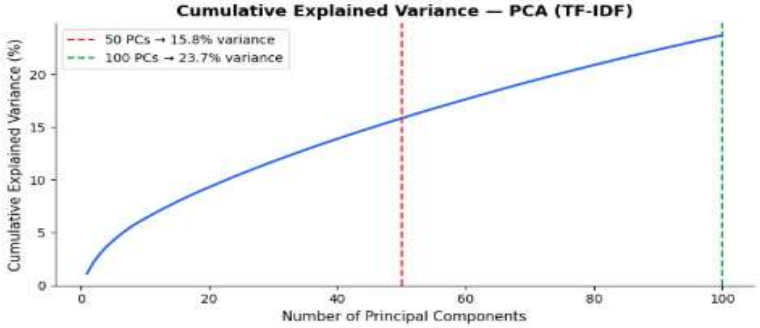


Figure 6.2.2 — PCA 2D Projection Coloured by True Sentiment Label

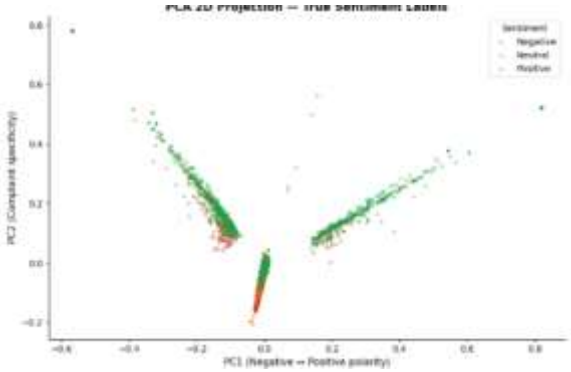
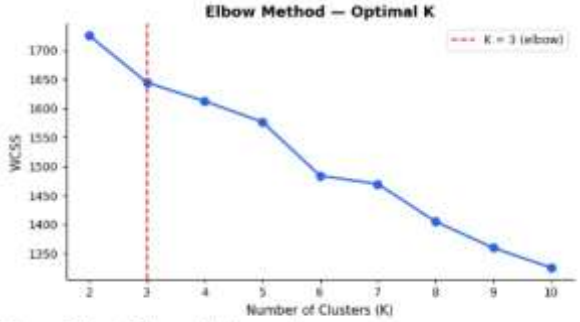


Figure 6.3.1 — Elbow Method: WCSS vs. Number of Clusters K




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Figure 8.1 – Elbow method saved.
Silhouette Score (K=3): 0.2179
Overall Cluster Purity: 0.6651
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Figure 6.3.2 — PCA 2D: True Sentiment Labels vs. K-Means Cluster Assignments

